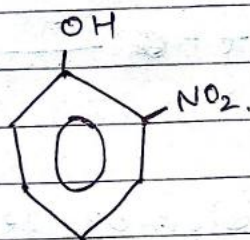


Chapter-4

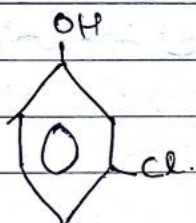
Phenol.



phenol.

O-nitrophenol.

2-nitrophenol.



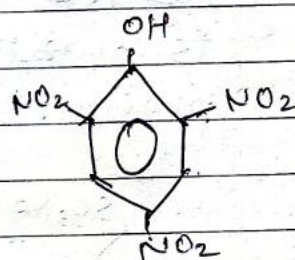
M-chlorophenol.

3-chlorophenol.



p-~~amino~~aminophenol.

4-aminophenol.

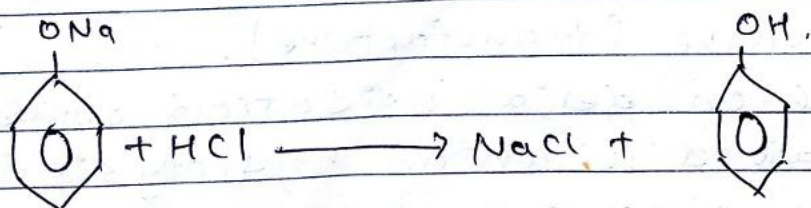
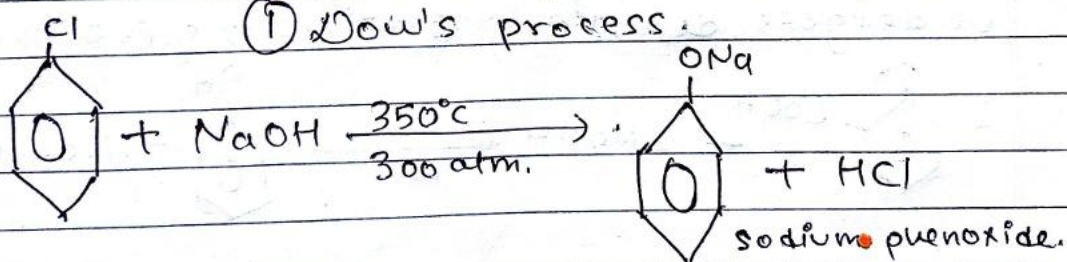


2,4,6-trinitrophenol.

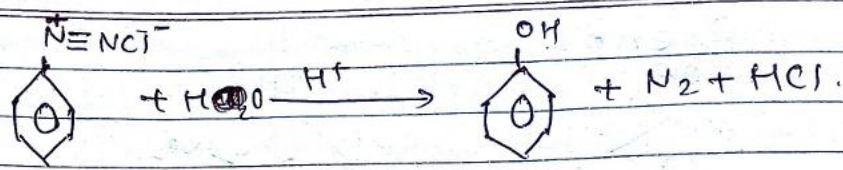
(picric acid).

General Methods of preparation of phenol.

(i) Dow's process.

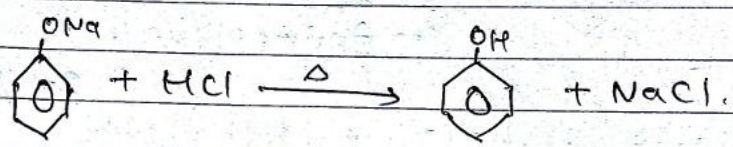
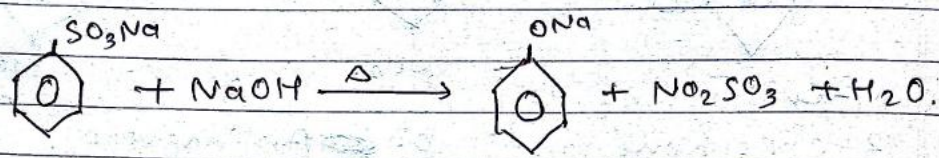


(ii) By hydrolysis of benzene diazonium chloride.



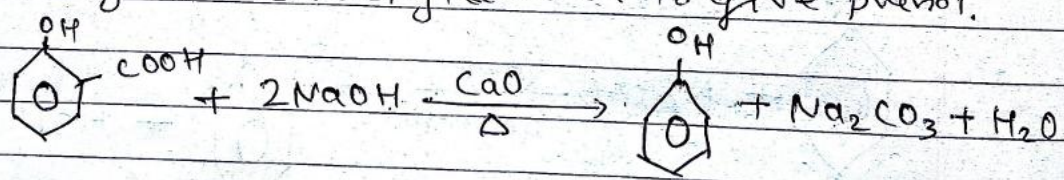
iii) from sodiumbenzene sulphonate.

→ Sodiumbenzene sulphonate when strongly heated with sodium hydroxide, sodium phenoxide is obtained which on acidification gives phenol.



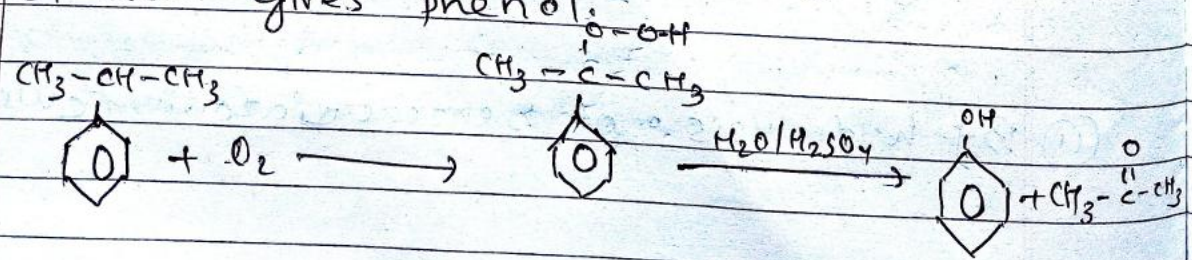
iv) From salicylic acid.

→ Salicylic acid when heated with soda lime undergoes decarboxylation to give phenol.



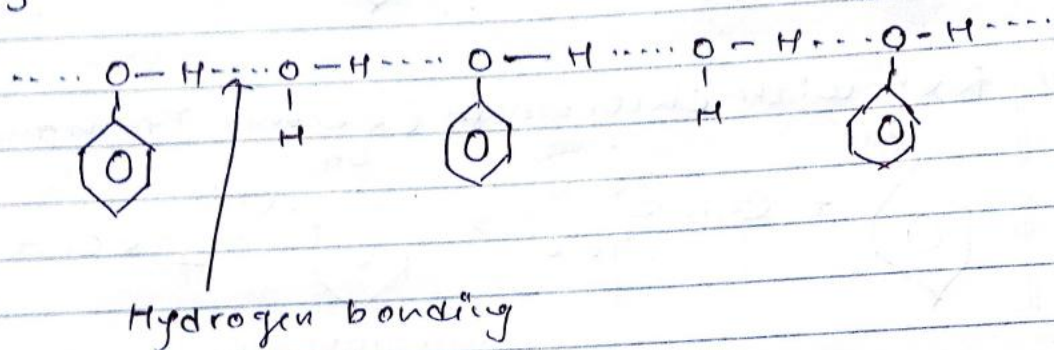
v) From Cumene (Manufacture).

→ Cumene on aerial oxidation, cumene peroxide is obtained which on hydrolysis in presence of acid gives phenol.



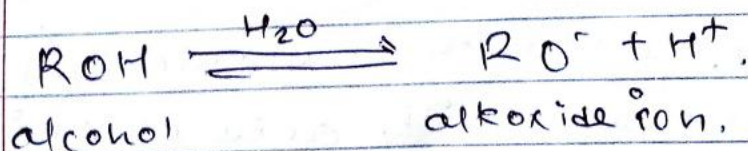
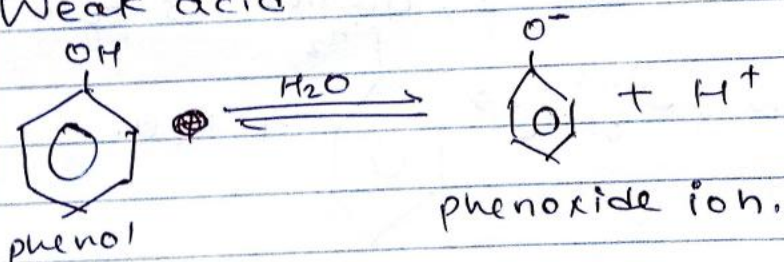
Physical properties of phenol.

- (i) Maybe solid or liquid.
- (ii) Melting point 38°C .
- (iii) Colourless but turns into brown when exposed to air.
- (iv) Odour = characteristic carbolic odour.
- (v) Slightly soluble in water because it can form hydrogen bond with water molecule.



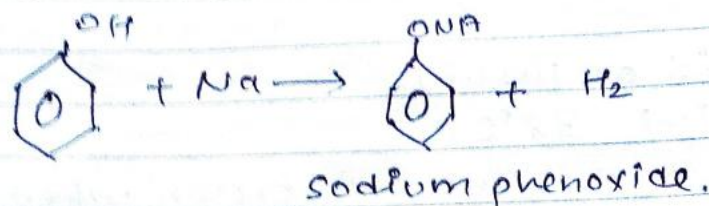
Chemical properties of phenol (Carbolic acid).

(i) Weak acid

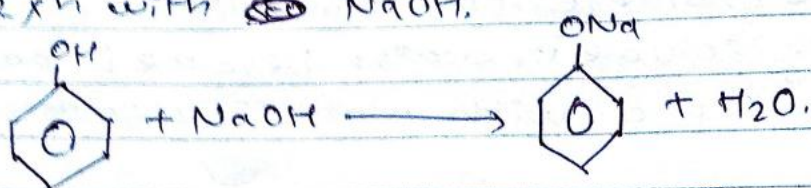


→ phenol is stronger acid than alcohol because phenoxide ion is more stable than alkoxide ion.

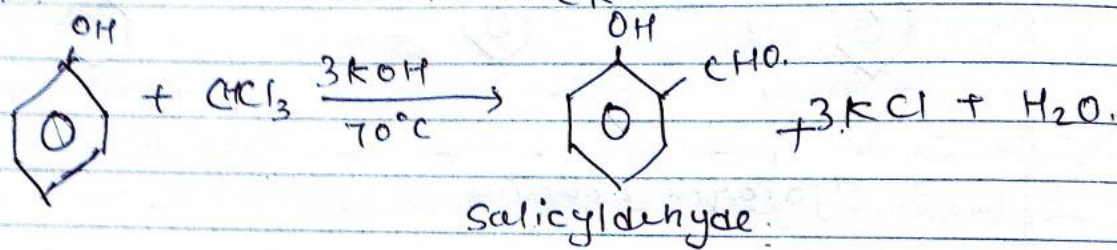
ii) Rxn with sodium.



iii) Rxn with ~~NaOH~~ NaOH.

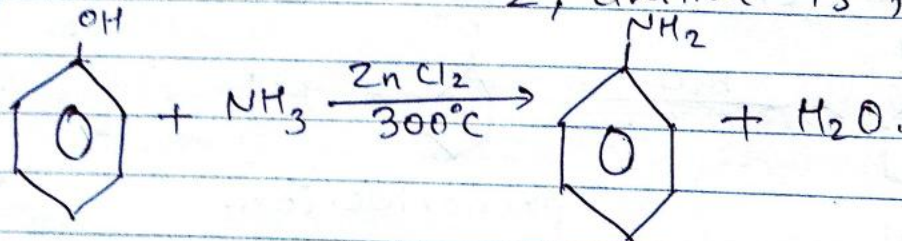


iv) Rxn with chloroform (Reimer Tiemann rxn)



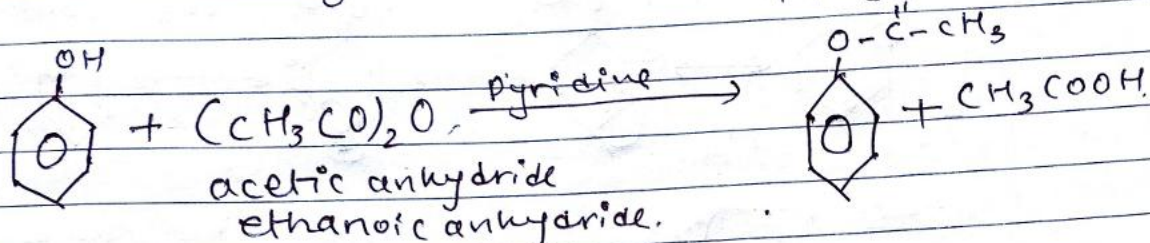
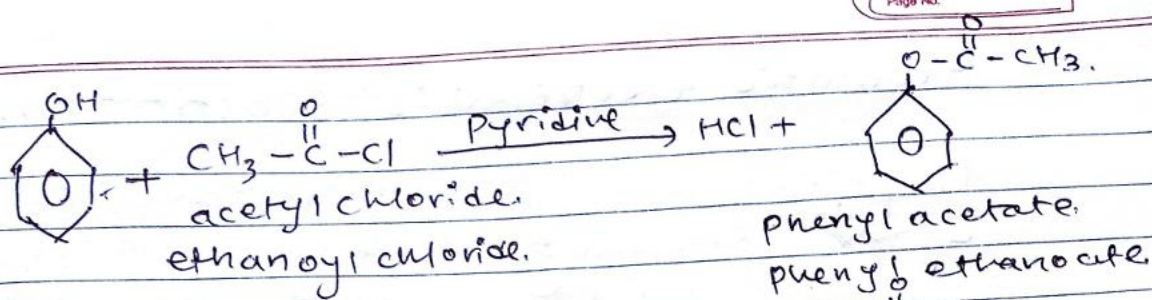
v) Rxn with Ammonia (Ammonolysis of phenol)

→ Phenol when strongly heated with NH_3 in presence of ZnCl_2 , aniline is formed.



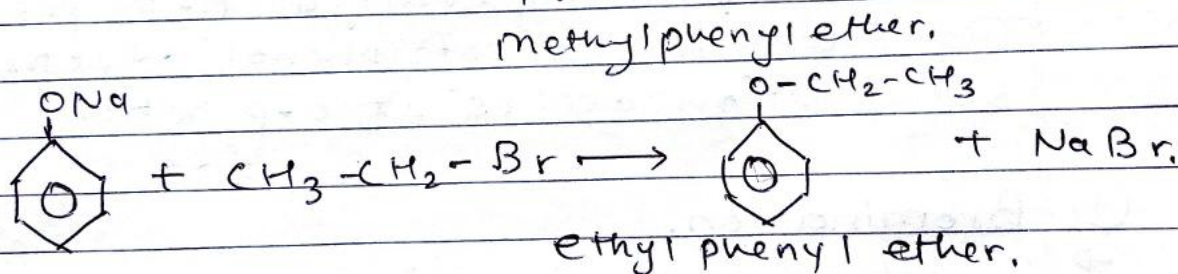
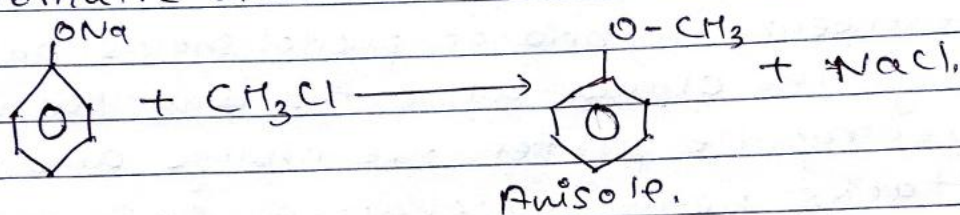
vi) ~~Phenol~~ Acylation, (Rxn with acid chloride)

→ Phenol when react with acid chloride or acid anhydride in presence of pyridine undergoes acylation at hydroxyl group to give aromatic ester.



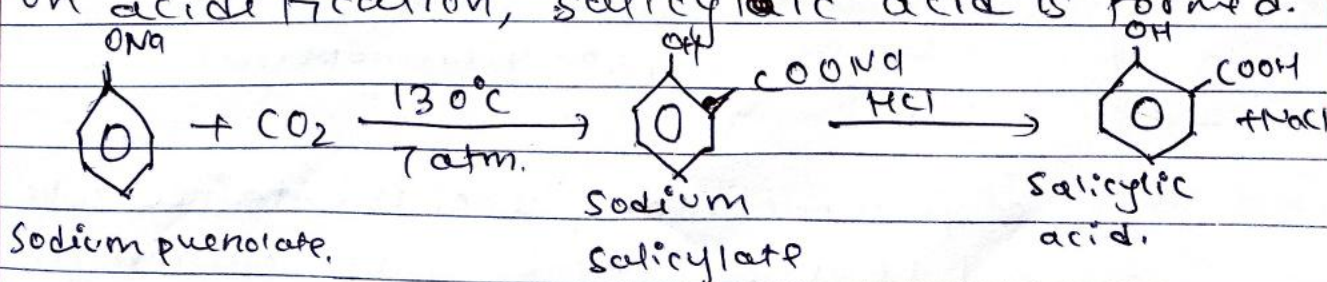
vii) Rxn with haloalkanes. (Williamson synthesis).

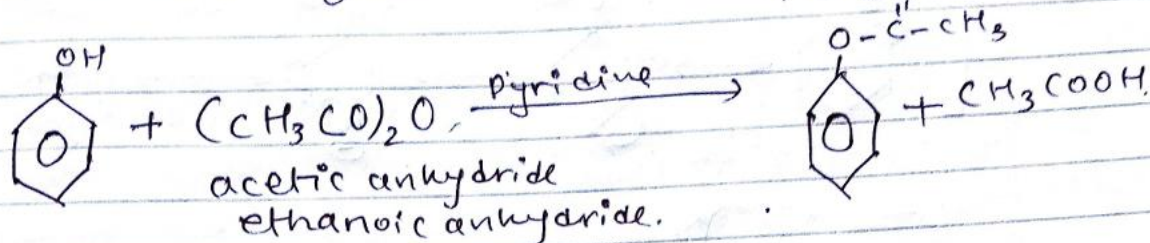
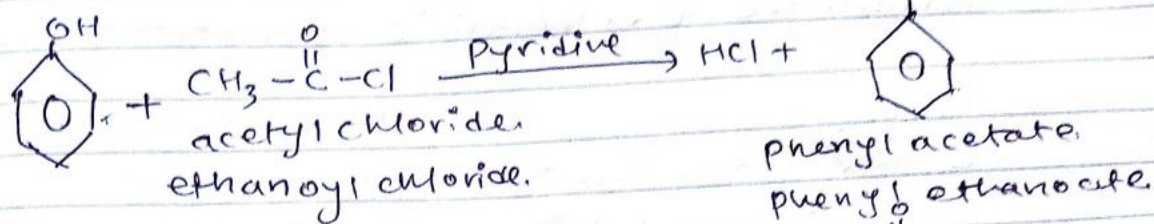
→ Sodium phenoxide when react with haloalkane, aromatic ether is obtained.



viii) Rxn with CO_2 . (Kolbe's rxn).

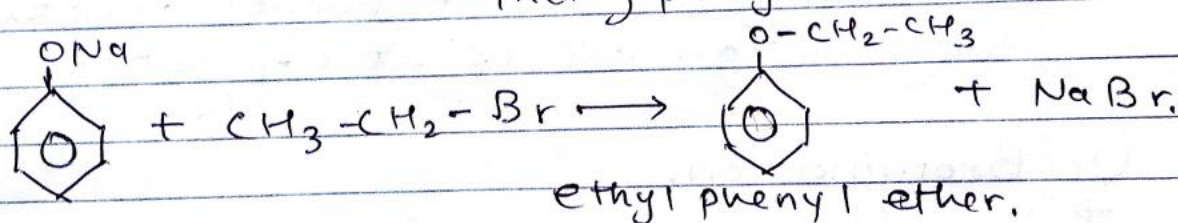
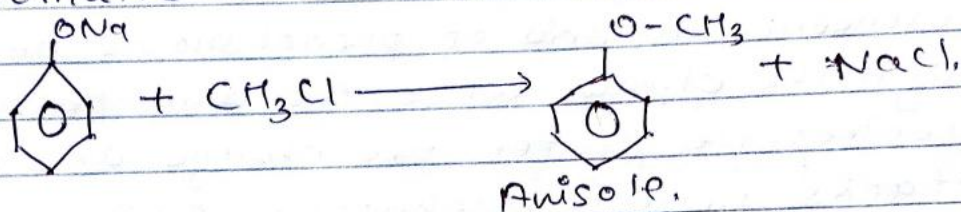
→ When sodium phenoxide is heated with CO_2 , under high pressure, sodium salicylate, which on acidification, salicylic acid is formed.





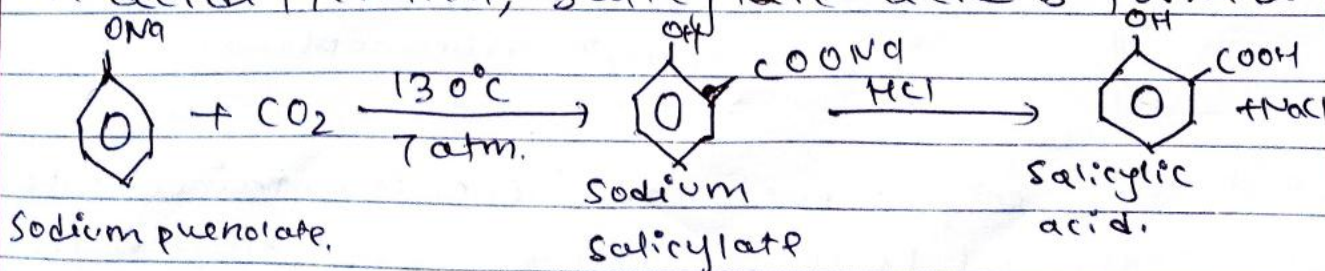
vii) Rxn with haloalkanes. (Williamson synthesis).

→ Sodium phenoxide when react with haloalkane, aromatic ether is obtained.



viii) Rxn with CO_2 . (Kolbe's rxn).

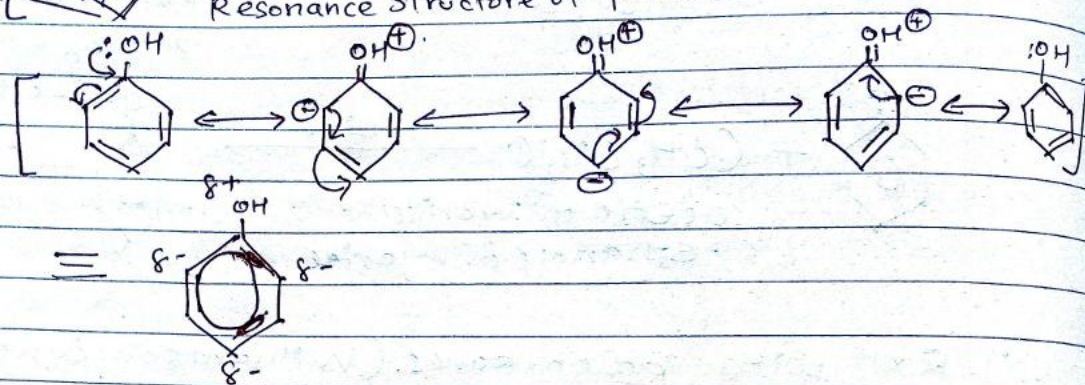
→ When sodium phenoxide is heated with CO_2 , under high pressure, sodium salicylate, which on acidification, salicylic acid is formed.



Aromatic electrophilic Substitution.



Resonance Structure of phenol.

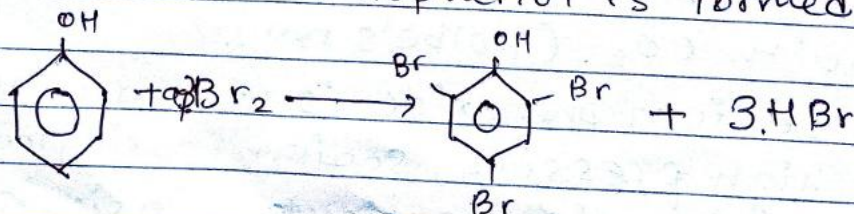


Resonance hybrid of phenol.

- Resonance hybrid of phenol shows partial negative charge at ortho and para position.
- Electrophile prefer -ve charge and attacks negatively charged O-P position. That's why in case of phenol, electrophilic Substitution occurs at o-p position.

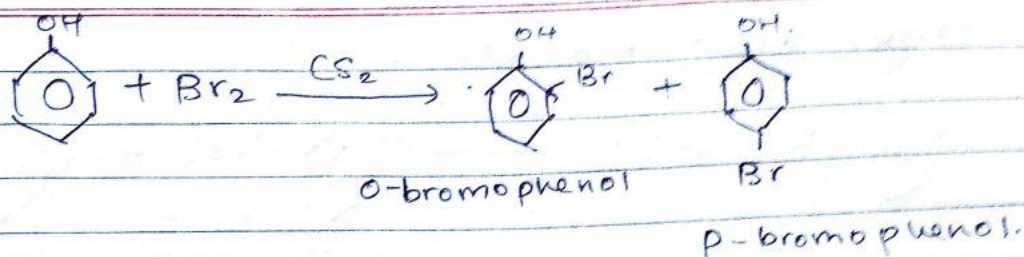
① Bromination.

- Phenol when treated with aqueous bromine 2,4,6 - tribromophenol is formed.



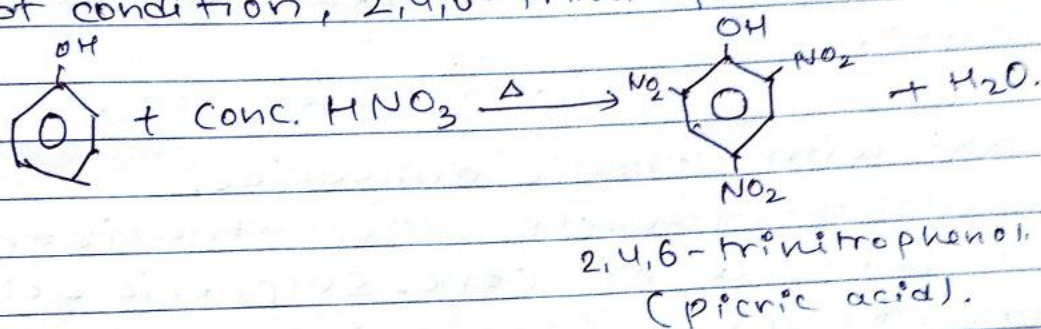
2,4,6 - tribromophenol.

- O or p monobromophenol is obtained when phenol reacts with bromine solution in carbon disulphide.

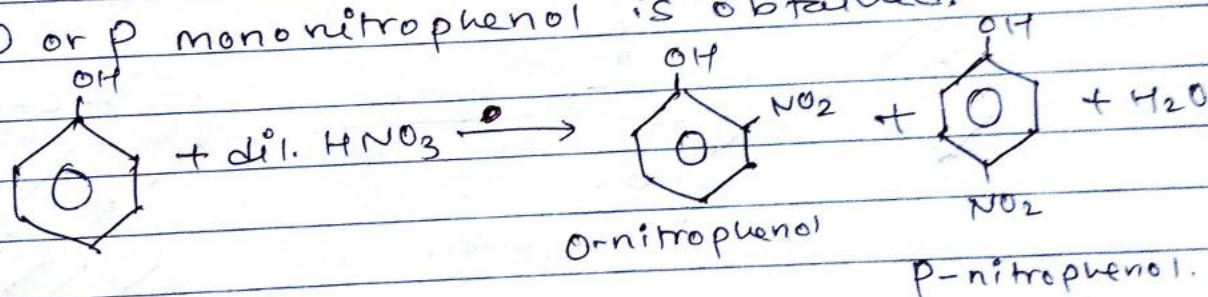


(ii) Nitration.

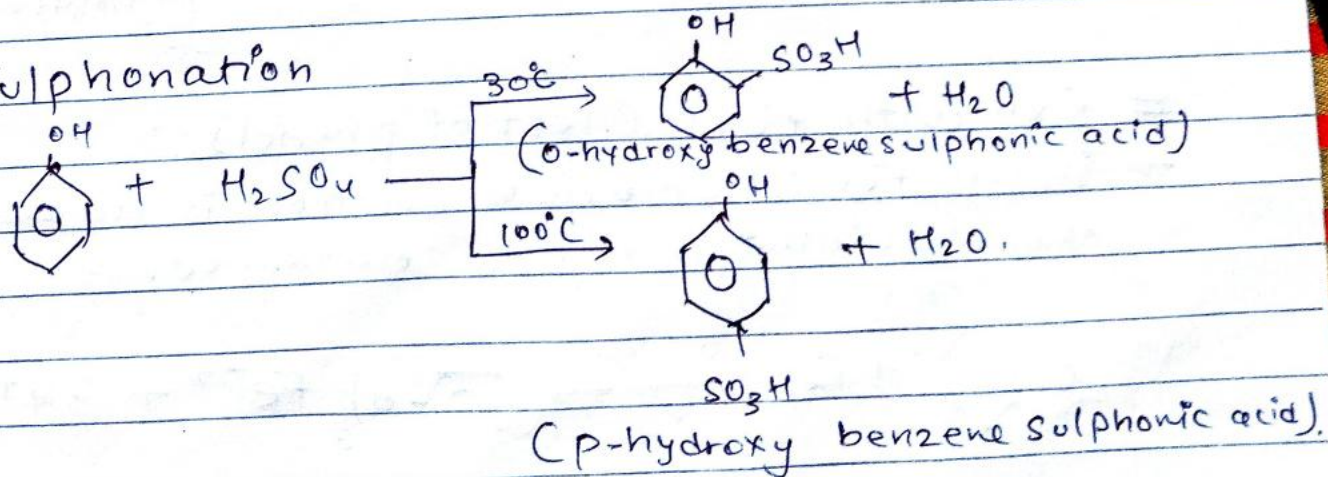
→ phenol when react with conc. Nitric Acid under hot condition, 2,4,6-trinitrophenol is obtained.



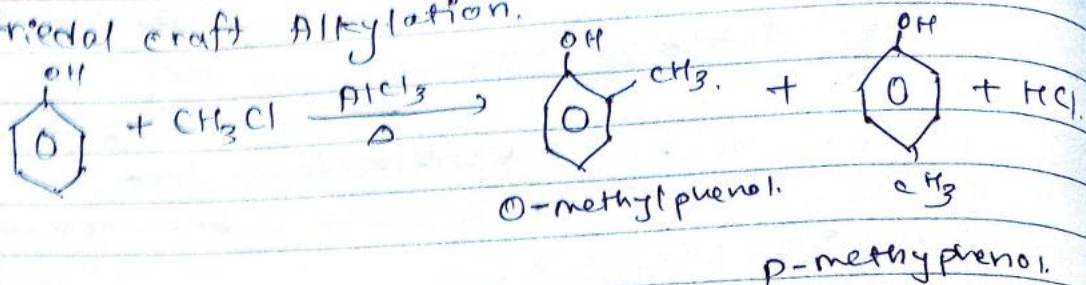
→ Phenol when react with dilute nitric acid o or p mononitrophenol is obtained.



(iii) Sulphonation



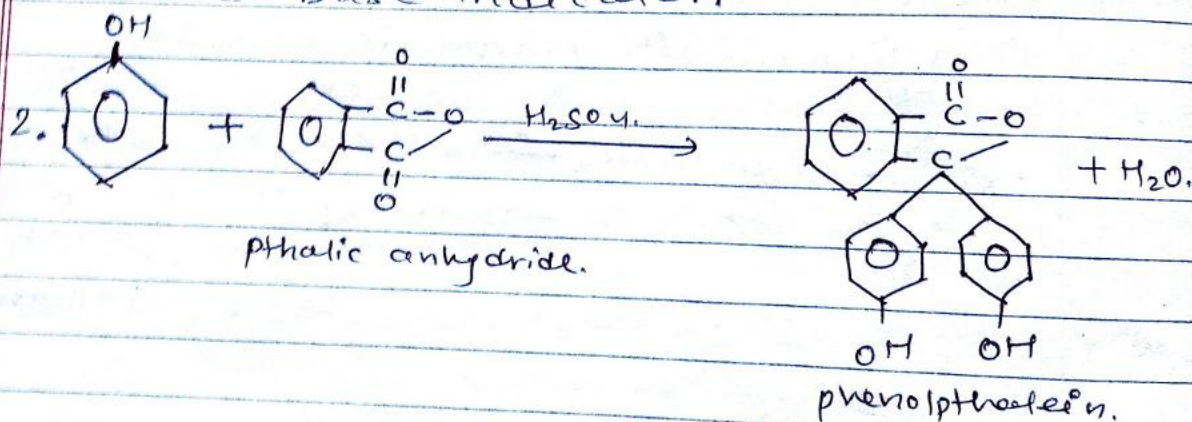
(iv) Friedel craft Alkylation.



(v) Phenol does not undergo Friedel craft acylation because in phenol, acylation occurs at hydroxyl group.

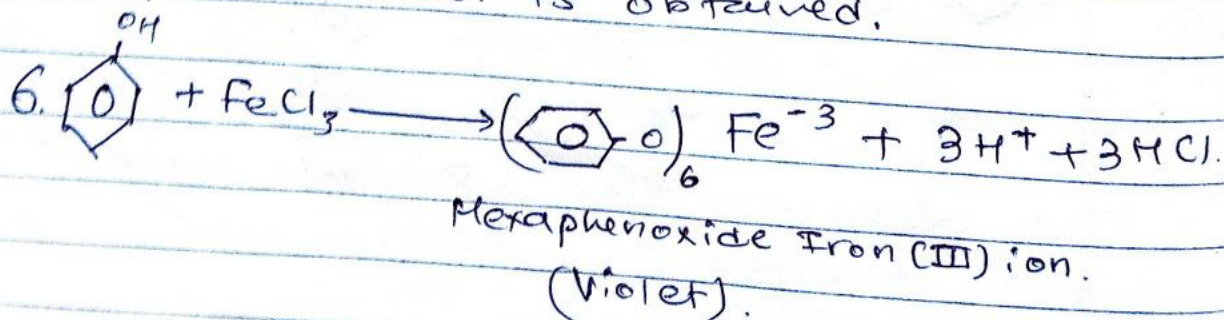
Rxn with phthalic anhydride.

→ Phenol when reacts with phthalic anhydride in presence of conc. sulphuric acid, phenolphthalein is obtained which is used as acid-base indicator.



Rxn with FeCl_3 (Test of phenol).

→ When Ferric chloride solution is added to phenol, violet colour is obtained.



Rxn with formaldehyde.

→ In acidic or basic medium, phenol and formaldehyde undergoes condensation to give highly cross linked thermosetting plastic called as bakelite.

